20 Here are the first five terms of a sequence.

−1 0 3 8 15

Find an expression, in terms of n, for the nth term of this sequence.

Terms: −1, 0, 3, 8, 15.

First differences: 1, 3, 5, 7; second difference 2.

Compare with n^2: 1, 4, 9, 16, 25.

Subtract 2n to obtain $n^2 - 2n$.

2 marks total
Second difference 2
$n^2 - 2n$

.....

(Total for Question 20 is 2 marks)

21 When a biased coin is thrown 4 times, the probability of getting 4 heads is $\frac{16}{81}$

Work out the probability of getting 4 tails when the coin is thrown 4 times.

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(Total for Question 21 is 2 marks)

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